

Multi-Species Audio Recognition: A CNN Approach for Biodiversity Monitoring

Jakob Jerše (#s1156064), Dariga Shokayeva (#s1158777), Daniel Duhnev (#s1158773)

1 Introduction

The BirdCLEF+ 2025 Kaggle Competition is a collaborative initiative aimed at advancing ecological restoration efforts through machine learning[1]. The challenge focused on automating species detection in soundscapes from Colombia’s critically endangered Magdalena Valley rainforests. The data provided contained a total of 28,564 labelled and 9,726 unlabelled audio files from a variety of 206 different species, including birds, amphibians, mammals, and insects. The goal was to build a multi-label classification model to identify which species are calling in 5-second audio clips in recordings made in El Silencio Natural Reserve, Colombia[2].

2 Method

2.1 Data analysis

The labelled training data consisted of recordings of various length from a variety of species each having its own species code. The unlabelled data included audio recordings from the same locations as the hidden test data. The unlabelled recordings could be used with a semi-supervised learning approach which could enhance model performance. Finally, the hidden test data included 700 one-minute 32 kHz recordings.

After manually inspecting the audio data and reviewing discussions [3] on the competition’s forum, we discovered that some labelled recordings contain human speech. In several cases, the human voice heavily dominated longer recordings, while the actual animal sound appeared only briefly, sometimes lasting as little as 1 second. The challenge was a multi-label classification problem with 206 potential species to predict. This meant the model needed to predict the probability of the presence of each species for each 5 second segment of each test recording. This meant that using 5 second slivers from the test recordings may be a useful strategy when training a model.

We decided to take a 3-step pipeline approach of first transforming the audio to Mel spectrograms[4], then training a CNN model[5], and finally running inference on the model[6]. This was inspired by publicly available notebooks present on the challenge. First we ran the pipeline end-to-end and used it as a baseline. Then we continued by first optimising the audio to Mel spectrogram transformation, followed by the model training. Finally, a separate notebook was used for inference, since upon inference we could not make use of the internet and were limited to using no more than 90 minutes of CPU time.

2.2 Data preprocessing

2.2.1 Mel spectrograms

After reviewing the literature and competition discussions [3] [7], we found that a reliable approach to tackle this challenge is to first convert audio files into Mel spectrograms[8]. A Mel spectrogram is a visual representation of audio that shows how energy in different frequency bands evolves over time, using the Mel scale, which reflects how humans perceive pitch. Unlike a regular spectrogram, it compresses higher frequencies logarithmically to emphasize features important to human hearing. By using Mel spectrograms, models can more effectively learn relevant audio patterns.

We used a sampling rate of 16 kHz to ensure compatibility with the Voice Activity Detection (VAD) tool used for speech removal described in the next subsection. Each spectrogram was computed with a window size of 512 samples and a hop length of 256, meaning each frame overlaps by 50%. We used 64 Mel bands to capture frequency content

from 50 Hz to 8 kHz, a range that includes most bird and animal vocalizations as well as human speech. The spectrogram was then converted to decibel (dB) scale for better dynamic range representation and then normalized to $[0, 1]$ to ensure training stability.

2.2.2 Speech removal

As stated in section 2.1, some recordings include human speech. To prevent models from learning speech patterns instead of focusing on animal sounds, we implemented a speech removal function. For that we utilize the pre-trained *Silero VAD* model[9]. This model works on audio sampled at 16 kHz and returns timestamps for segments it classifies as speech. We used a high confidence threshold of 0.9 to ensure that no animal sound data was removed. To avoid cutting too close to animal sounds, we also added 100 ms of padding before and after each detected segment. All detected speech parts were removed, and if no speech was found, the audio was left unchanged. It is also important to note that although this method may remove overlapping animal sounds when they occur simultaneously with human speech, it helps ensure that only the clearest and most distinct vocalizations are retained. We designed and ran experiments against a baseline model to test this hypothesis.

2.2.3 Silence trimming

Additionally, we explored two methods for handling silent segments, which typically do not provide useful information for training. The first method removed all audio below a certain amplitude threshold, resulting in a more aggressive silence removal. However, this approach also cut out short pauses between vocalizations, making the audio sound unnaturally fast and disrupting its natural rhythm. To address this, we implemented a second method that identifies non-silent intervals using a 30 dB threshold and removes only silences longer than 1 second between these intervals. Shorter silences are preserved to maintain a realistic acoustic context, helping the model learn temporal patterns in animal vocalizations.

2.2.4 Whole preprocessing pipeline

The preprocessing pipeline starts by loading the audio and resampling it to 16 kHz to ensure compatibility with the speech removal model. Human speech is removed using a confidence threshold of 0.9, followed by the removal of silences longer than 1 second based on a 30 dB threshold. If the resulting audio is shorter than the target duration of 5 seconds, it is repeated rather than padded with silence to reach the desired length. We hypothesised that repetition might be preferable to silence padding, as it adds content, even if redundant, whereas silence contributes no additional information. If the resulting audio is longer than 5 seconds, a random 5 second segment is extracted. Finally, a Mel spectrogram is generated from each processed audio.

2.3 Model selection and Training

2.3.1 Baseline notebook training setup

There is substantial literature supporting the use of CNN models using Mel spectrograms for audio detection tasks[10][11]. As a starting point, we chose a baseline notebook[12] which trains an ensemble of five EfficientNet-B0[13][14] models using 5-fold cross-validation. Each fold is trained for 10 epochs, and the best performing model in each fold is selected based on the highest AUC score. The models are trained on Mel spectrogram inputs generated during preprocessing. To improve robustness and generalization, spectrogram augmentations such as time masking, frequency masking, and brightness/contrast adjustments are applied during training.

2.3.2 Model Choice and Model Refinement

After optimising the data preprocessing, we designed experiments to test and improve on the model training process. After considering the literature on the utility of different CNN model architectures[15] we found that NFNet models often perform well in image-based tasks, achieving comparable accuracy to EfficientNet models, while having much faster train time. We experimented with ECA-NFNet-L0, however, we found that a 5-fold CV of 10 epochs each experienced timeouts due to the size of the model. Instead, we experimented with single fold and 3-fold cross-validation models and compared them to the baseline. We extended this setup by modifying the optimizer and scheduler while increasing the number of training epochs to examine the effect of optimization strategy on performance.

We also experimented with training a single EfficientNet-B0 model without cross-validation, based on flash talks and discussion forums[16], where other competitors reported that it showed little or no improvement in their results. In

addition, we explored ensembling ECA-NFNet-L0 and ECA-NFNet-L1, without using cross-validation, to evaluate whether a combination of differently sized NFNet variants could perform well as a lightweight ensemble.

2.3.3 Pseudo-Labelling

To use the unlabelled soundscape recordings, we generated Mel spectrograms for each 5-second segment using our optimal preprocessing pipeline. We then applied our 5-fold EfficientNet-B0 ensemble to predict labels. Segments whose top-class probability exceeded a confidence threshold were added as pseudo-labels. We merged these with the original labelled data and retrained our CNN from scratch for submission.

Using a fixed threshold of 0.9 yielded only 2,200 pseudo-labels (of 9,700) and degraded performance, likely amplifying abundant classes while neglecting rare species. Instead, we adopted a non-linear threshold inversely proportional to class frequency, boosting rare-species examples[17]. This added 6,200 high-confidence samples and improved our final model.

3 Results

3.1 Baseline Modifications & LB Scores

Table 1 presents the leaderboard (LB) scores achieved across all experiments. The preprocessing experiments show progressive performance improvements, with individual techniques (random clipping, speech removal, silence removal) yielding modest gains, while their combination significantly enhanced performance. The final optimised preprocessing pipeline achieved an LB score of 0.810. From the training modification that proceeded, pseudo-labelling was the strategy that delivered the highest improved performance on the public LB with a score of 0.816[18], representing a substantial improvement over the baseline score of 0.761. On the other hand, the 3 model NFNet ensemble achieved the highest on the private LB with a score of 0.814[19]. Due to the page limit of the report, we have not included all experiments. Some experiments included minor changes which led to worse performance, and others resulted in time-out issues due to Kaggle compute limitations.

3.2 Score Variability Analysis

Random seed selection demonstrated considerable impact on LB scores throughout the experiments as shown in Table 2. To achieve reproducibility, a constant random seed was chosen and was kept for future experiments during model training.

3.3 Optimal Configuration

The highest-performing configuration based on public LB combined: (1) speech removal and silence trimming with conservative thresholds[20], (2) random 5-second clip extraction[20], (3) the addition to pseudo-labelled data during training[18], and (4) consistent random seed for reproducibility. This configuration achieved a public LB score of 0.816 and a private LB score of 0.809, demonstrating substantial improvements of 7.2% and 8.2% over the baseline performance on the public and private leaderboards, respectively. For the private leaderboard, 2 submissions (shown in *Italic* in Table 1) were pre-selected by us, with the highest achieved private LB score of 0.812. The best notebook based on the private leaderboard, achieving a score of 0.814, involved a final preprocessing pipeline and ensembling of 3 `eca_nfnet_l0` models[19].

4 Discussion

The results highlight that the most significant improvements in leaderboard scores were achieved through a carefully designed preprocessing pipeline rather than through pseudo-labelling or model ensembling alone. Specifically, the combination of speech removal, conservative silence trimming, and random 5-second clip extraction led to the highest improvements in LB scores, demonstrating the critical importance of addressing noise and irrelevant segments in ecological audio data. While pseudo-labelling and ensembling strategies did yield notable improvements compared to the baseline, their impact was slightly less pronounced than that of optimized preprocessing, particularly on the public leaderboard. On the private leaderboard, the best performance (0.814) was obtained by ensembling three ECA-NFNet-L0 models using the final preprocessing pipeline, highlighting the complementary value of both robust

Table 1: Summary of Notebook Changes and LB Scores

Change Type	Public LB Score	Private LB Score	Explanation
Baseline	0.761	0.748	Ran 3-step baseline pipeline.
Audio to Mel Spectrogram Preprocessing			
Speech removal	0.788	0.796	Add speech removal with 0.9 confidence threshold.
Silence removal	0.786	0.793	Remove any silence >1s.
Speech and silence removal	0.783	0.796	Combine speech and silence removal.
Random audio clipping	0.779	0.786	Choose random 5s from each audio instead of central 5s used in baseline.
Speech removal + random 5s clip	0.802	0.794	Combine speech removal and random 5s.
Speech & silence removal + random clip	0.806	0.782	Combine all 3 changes.
Random 1s clips concatenation	0.798	0.778	Combine all 3 changes but choose 5 random 1s clips and concatenate them.
<i>Final preprocessing</i> [20]	0.810	0.812	Combine all 3 changes. Fixed random seed (arbitrary but constant).
Model Training			
1 model, no CV	0.785	0.782	Train 1 model with no cross-validation instead of 5 model ensemble.
Switch to nfnet, single model	0.782	0.785	Used eca_nfnet_l0 single model.
Ensemble of eca_nfnet_l0 & eca_nfnet_l1	0.790	0.795	Train and ensemble 1 eca_nfnet_l0 and 1 eca_nfnet_l1 models.
Ensemble of 3 eca_nfnet_l0 models[19]	0.804	0.814	
<i>Pseudo-labelling with "Final Preprocessing" model</i> [18]	0.816	0.809	Pseudo-labelled dataset added in training with best performing model.

Table 2: Grouped Experiments and Observed LB Scores

Change	Public LB	Seeds	Notes
Baseline	0.761, 0.796	N/A	$\Delta = 0.035$ across identical runs
Best preprocessing	0.819, 0.797	N/A	$\Delta = -0.022$ across identical runs
Final preprocessing	0.806, 0.794, 0.810	N/A, 42, 20	$\Delta = \pm 0.016$ with different seeds

data preparation and model aggregation.

These findings suggest that, in the context of complex soundscape classification, rigorous preprocessing to remove confounding audio artifacts, such as human speech and long silences, can have an even greater effect on model accuracy than more advanced training or inference techniques. Additionally, the observed variability in leaderboard scores due to random seed selection emphasizes the need for consistent experimental protocols and reproducibility. Overall, the project demonstrates that a strong foundation of data cleaning and augmentation is essential for maximizing the effectiveness of modern deep learning models in challenging, real-world bioacoustic applications.

5 Conclusion

Our results for the BirdCLEF+ 2025 competition show that the biggest gains in multi-species audio recognition came from a rigorous preprocessing pipeline—combining speech removal, silence trimming, and random 5-second segment extraction. This data cleaning approach had a greater impact than pseudo-labeling or model ensembling. The highest public leaderboard score (0.816) came from this preprocessing combined with pseudo-labeling, while the top private score (0.814) used an ensemble of three ECA-NFNet-L0 models with the same pipeline. These results highlight the critical role of targeted preprocessing over complex training strategies and the importance of experimental consistency.

6 Code Notebooks

Please find the list of shared notebooks referenced in the report below:

- Final data preprocessing [20]
- Final training code [21]
- Final inference code [22]
- NFNet 3-fold CV model ensemble [19]
- Final code for pseudo-labelled dataset generation [17]
- Final training code with pseudo-labelling [18]

7 Generative A.I. Acknowledgement

During the preparation of this report, we used generative AI (ChatGPT) to proofread paragraphs and suggest improvements to enhance readability. All suggestions were reviewed and edited as necessary. We take full responsibility for the final content of the report.

8 Author Contributions

The work on the project and report we feel was divided equally, and working mostly in person really helped with that. In Table 3, we present a summary of each author’s contributions.

Dariga	Conducted a literature review and a review of the public discussions and notebooks and contributed initial ideas for experiments. Explored different model architectures and worked to refine models via hyperparameter tuning. Worked on writing and editing multiple sections in the report.
Daniel	Ran the 3 step pipeline end-to-end to produce a baseline result. Conducted an initial data analysis and contributed ideas for experiments. Implemented pseudo-labelling for the unlabelled data. Worked on writing and editing multiple sections in the report.
Jakob	Explored silence trimming and voice removal with different parameters and thresholds. Conducted an initial data analysis and contributed ideas for experiments. Explored different model architectures and worked to refine models via hyperparameter tuning. Worked on writing and editing multiple sections in the report.

Table 3: Author Contributions

9 Evaluation of the Process

Overall, we thought the project went pretty smoothly, similar to the first one. We held regular team meetings, meeting in person at least once a week to share updates on our progress and discuss the results we achieved. We preferred working together in the same room, but sometimes worked from home being on a call or working individually on our assigned tasks. The work was divided effectively, and after each meeting, we planned the next steps and determined what to try next.

9.1 What went well

Similar to the first project, we believe we had a good team dynamic. Our regular team meetings ensured that everyone was up to date on the current status of the project, helping us maintain alignment. Additionally, working together in person was beneficial, as it allowed us to address issues or questions immediately, rather than waiting for the next meeting or a response in the group chat. We also aimed to complete our assessment forms with the course’s

supervisors as soon as possible to discuss our current challenges and ideas and get useful feedback early.

A lesson learned and an improvement from the first project was our better planned scientific approach and our extensive experimentation when testing different changes. We were more methodological, first focusing on and optimising the preprocessing step, followed by the model training and refinement. We aimed to make a lot of experiments and test many different parameters. This meant that we could pick the experiments we thought had added the most value to our discussion within the report's page limit. It should be noted that there were other smaller experiments that we performed when optimising our model which are not included in the report due to the page limit. Finally, similar to the first project, we learned a lot about Machine Learning in practice from doing this competition and we hope this is also evident from the report we produced.

9.2 What could have been improved

While being satisfied with the challenge overall, there are areas where we could have been better. For example, we chose a baseline model earlier on in the competition and identified experiments we wanted to explore. This meant that we focused on working our way up from that particular baseline and we fell back on the most recent public discussions and notebooks on the challenge. We believe that we could have produced a better final model had we chosen a better baseline or kept up to date earlier with the latest discussions, but due to the limited time we could spent on the competition this was not feasible. We do feel, however, that we learned a lot and that we explored sufficient ideas to produce a significant overall improvement in our final model.

10 Evaluation of the Supervision

Similar to the first project, we attended the tutorial sessions and did our check in forms early, which we found valuable. The check-in forms in particular were helpful, as discussing our responses with the TA reassured us that we were on the right track and that our plan for future experiments was on point. We also liked the flash talks, where we heard about our colleagues' approaches, giving us ideas to implement ourselves.

References

- [1] H. Klinck, J. S. Cañas, M. Demkin, S. Dane, S. Kahl, and T. Denton, "Birdclef+ 2025," <https://kaggle.com/competitions/birdclef-2025>, 2025, kaggle.
- [2] Fundación Biodiversa Colombia, "El silencio natural reserve, colombia," <https://www.fundacionbiodiversa.org/fundacion2024/es/information/>, 2024, accessed: 2025-06-05.
- [3] K. Dmitrev. (2025) Birdclef 2025 discussion: Cross-validation insights. Accessed: 2025-06-07. [Online]. Available: <https://www.kaggle.com/competitions/birdclef-2025/discussion/568886>
- [4] K. İdrisoğlu, "Transforming Audio-to-Mel Spec. — BirdCLEF'25: BirdCLEF 2025 Data Preprocessing Notebook," <https://www.kaggle.com/code/kadircandrisolu/transforming-audio-to-mel-spec-birdclef-25>, 2025, kaggle Notebook; Accessed: 2025-06-08.
- [5] —, "EfficientNet B0 Pytorch [Train] — BirdCLEF'25," <https://www.kaggle.com/code/kadircandrisolu/efficientnet-b0-pytorch-train-birdclef-25>, 2025, kaggle Notebook; Accessed: 2025-06-08.
- [6] —, "EfficientNet B0 Pytorch [Inference] — BirdCLEF'25," <https://www.kaggle.com/code/kadircandrisolu/efficientnet-b0-pytorch-inference-birdclef-25>, 2025, kaggle Notebook; Accessed: 2025-06-08.
- [7] S. Ahmed. (2025) Birdclef 2025 discussion: [insert discussion title here]. Accessed: 2025-06-07. [Online]. Available: <https://www.kaggle.com/competitions/birdclef-2025/discussion/573066>
- [8] L. Roberts. (2020) Understanding the mel spectrogram. Accessed: 2025-06-07. [Online]. Available: <https://medium.com/analytics-vidhya/understanding-the-mel-spectrogram-fca2afa2ce53>
- [9] S. Team. (2021) Silerovad: Machine learning model to detect speech segments. Accessed: 2025-06-07. [Online]. Available: <https://medium.com/axinc-ai/silerovad-machine-learning-model-to-detect-speech-segments-e99722c0dd41>

- [10] M. Huzaifah, “Comparison of time-frequency representations for environmental sound classification using convolutional neural networks,” *arXiv preprint arXiv:1706.07156*, Jun. 2017. [Online]. Available: <https://arxiv.org/abs/1706.07156>
- [11] H. Purwins, B. Li, T. Virtanen, J. Schlüter, S.-Y. Chang, and T. Sainath, “Deep learning for audio signal processing,” *arXiv preprint arXiv:1905.00078*, May 2019. [Online]. Available: <https://arxiv.org/abs/1905.00078>
- [12] K. İdrisoğlu. (2025) Efficientnet b0 pytorch [train] — birdclef’25. Accessed: 2025-06-07. [Online]. Available: <https://www.kaggle.com/code/kadircandrisolu/efficientnet-b0-pytorch-train-birdclef-25>
- [13] M. Tan and Q. V. Le, “Efficientnet: Rethinking model scaling for convolutional neural networks,” in *Proceedings of the 36th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, K. Chaudhuri and R. Salakhutdinov, Eds., vol. 97. PMLR, 2019, pp. 6105–6114. [Online]. Available: <https://proceedings.mlr.press/v97/tan19a.html>
- [14] S. Herath, “EfficientNetB0 architecture — Stem Layer,” <https://medium.com/image-processing-with-python/efficientnetb0-architecture-stem-layer-496c7911a62d>, Jan. 2024, medium.
- [15] K. Wu. (2021) Nfnets explained: Deepmind’s new state-of-the-art image classifier. Accessed: 2025-06-07. [Online]. Available: <https://medium.com/data-science/nfnets-explained-deepminds-new-state-of-the-art-image-classifier-10430c8599ee>
- [16] S. Ahmed. (2025) Birdclef 2025 discussion: [insert discussion title here]. Accessed: 2025-06-07. [Online]. Available: <https://www.kaggle.com/competitions/birdclef-2025/discussion/570402>
- [17] J. Jerse, D. Shokayeva, and D. Duhnev. (2025) Pseudo-label creator — mlip 33 19725e. Accessed: 2025-06-09. [Online]. Available: <https://www.kaggle.com/code/jakobjere/pseudo-label-creator-mlip-33>
- [18] —. (2025) Efficientnet b0 pytorch [train] — labelled+pseudo. Accessed: 2025-06-09. [Online]. Available: <https://www.kaggle.com/code/daniduhnev/efficientnet-b0-pytorch-train-labelled-pseudo>
- [19] —. (2025) Efficientnet b0 pytorch [train] — nfnet 3 model. Accessed: 2025-06-09. [Online]. Available: <https://www.kaggle.com/code/daniduhnev/efficientnet-b0-pytorch-train-nfnet-3-model/notebook>
- [20] —. (2025) Audio-to-mel spec. — birdclef’25 mlip 33 19725e. Accessed: 2025-06-09. [Online]. Available: <https://www.kaggle.com/code/darigashokayeva/audio-to-mel-spec-birdclef-25-mlip-33-19725e>
- [21] —. (2025) Training — birdclef’25 mlip 33. Accessed: 2025-06-09. [Online]. Available: <https://www.kaggle.com/code/darigashokayeva/training-birdclef-25-mlip-33/notebook?scriptVersionId=244257667>
- [22] —. (2025) Inference — birdclef’25 mlip 33. Accessed: 2025-06-09. [Online]. Available: <https://www.kaggle.com/code/darigashokayeva/inference-birdclef-25-mlip-33?scriptVersionId=244257313>